CIE International A Level Physics

# Topic 20

# Magnetic Fields

Revision Booklet

**This booklet covers:**

- Magnetic Fields and Field Lines
- Force on a Current-Carrying Conductor
- Force on a Moving Charge
- Hall Effect and Velocity Selector
- Magnetic Fields due to Currents
- Electromagnetic Induction

# Magnetic Fields

### Magnetic Field

A **magnetic field** is a region of space in which a moving charge, or a current-carrying conductor, experiences a force.

- Magnetic fields are produced by **moving charges** (electric currents) or by **permanent magnets**.
- The field is represented by **field lines** (flux lines); the direction is the direction of the force on a north pole.
- Field lines never cross; closer lines indicate a stronger field.
- Crosses ($\times$) represent field into the page; dots ($\bullet$) represent field out of the page.

# Force on a Current-Carrying Conductor

### Force on a Conductor

$$F = BIL\sin\theta$$

$F$ = force on the conductor (N)
$B$ = magnetic flux density (T)
$I$ = current in the conductor (A)
$L$ = length of conductor in the field (m)
$\theta$ = angle between the conductor and the field direction

Force is **maximum** when $\theta = 90°$ (conductor perpendicular to field): $F = BIL$.
Force is **zero** when $\theta = 0°$ (conductor parallel to field).

### Magnetic Flux Density $B$

**Magnetic flux density** is defined as the force per unit current per unit length acting on a wire placed **at right angles** to the field.

$$B = \frac{F}{IL} \qquad \text{units: T (tesla)} \equiv \text{N A}^{-1}\text{m}^{-1}$$

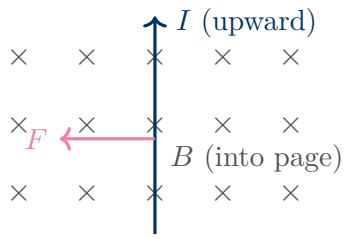
$B$ is a vector quantity. 1 T is a strong field; Earth's field is $\approx 50\ \mu$T.

### Fleming's Left-Hand Rule

For a **conventional current** in a magnetic field:

- **Thumb**: direction of force (motion) $F$
- **First finger**: direction of magnetic field $B$
- **Second finger**: direction of conventional current $I$

Remember: the rule gives the force on *positive* charges / conventional current. For electrons, reverse the direction.




## Force on a Moving Charge

### Force on a Moving Charge

$$F = BQv \sin\theta$$

$F$ = magnetic force (N)
$B$ = magnetic flux density (T)
$Q$ = charge (C)
$v$ = speed of the charge (m s$^{-1}$)
$\theta$ = angle between velocity and field

For $\theta = 90°$: $F = BQv$, directed perpendicular to both $v$ and $B$.

### Circular Motion in a Magnetic Field

When a charged particle moves **perpendicular** to a uniform magnetic field, the magnetic force is always perpendicular to the velocity, so no work is done and the **speed is constant**. The particle moves in a **circle**:

$$BQv = \frac{mv^2}{r} \quad \Rightarrow \quad r = \frac{mv}{BQ}$$

A larger momentum or smaller $B$ gives a larger radius.

## The Hall Effect

### Hall Voltage

When a current-carrying conductor is placed in a magnetic field perpendicular to the current, charge carriers experience a sideways force. They accumulate on one face until the electric force balances the magnetic force, producing the **Hall voltage**:

$$V_H = \frac{BI}{ntq}$$

$V_H$ = Hall voltage (V)
$B$ = magnetic flux density (T)
$I$ = current through the conductor (A)
$n$ = number density of charge carriers ($m^{-3}$)
$t$ = thickness of the conductor in the direction of $B$ (m)
$q$ = charge on each carrier (C)

A **Hall probe** uses this effect to measure $B$: since $V_H \propto B$ at constant $I$.

## Velocity Selector

### Velocity Selector

A velocity selector uses **crossed** electric and magnetic fields so that only particles with a specific speed pass through undeflected:

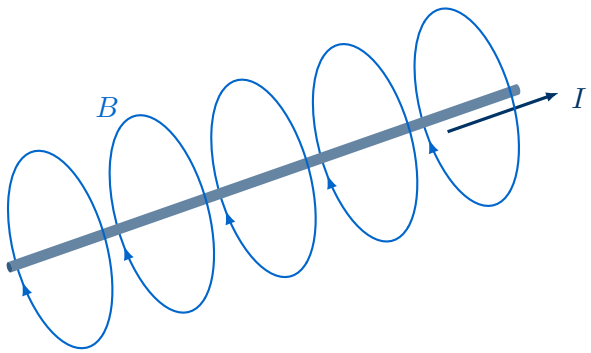
$$qE = BQv \quad \Rightarrow \quad v = \frac{E}{B}$$

- Electric force $qE$ and magnetic force $BQv$ act in **opposite** directions.
- Only particles where these forces balance travel in a straight line.
- Faster particles are deflected one way; slower particles the other.

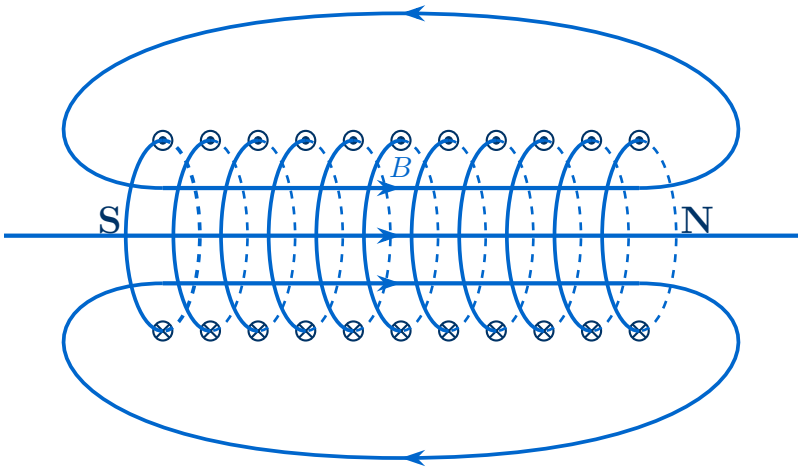
## Magnetic Fields due to Currents

### Field Patterns

- **Long straight wire**: concentric circles centred on the wire. Direction given by the **right-hand grip rule** — thumb points in direction of current, fingers curl in direction of field.
- **Flat circular coil**: field lines pass through the centre of the coil. Field at the centre is approximately uniform over a small region.
- **Long solenoid**: nearly uniform field inside, similar to a bar magnet outside. A **ferrous (iron) core** greatly increases the field strength.




**Magnetic field around a straight wire**




**Magnetic field around a Solenoid**

### Forces Between Parallel Conductors

- **Same direction currents**: conductors **attract** each other.
- **Opposite direction currents**: conductors **repel** each other.
- Each conductor sits in the magnetic field produced by the other; the force is given by $F = BIL$.

## Electromagnetic Induction

### Magnetic Flux

**Magnetic flux** $\Phi$ is the product of the magnetic flux density and the cross-sectional area perpendicular to the field:

$$\Phi = BA\cos\theta \qquad \text{units: Wb (weber)} \equiv \text{T m}^2$$

When $B$ is perpendicular to the area: $\Phi = BA$.

**Flux linkage** $N\Phi$ is the flux through a single turn multiplied by the number of turns $N$ of the coil: units Wb-turns.

### Faraday's Law and Lenz's Law

$$\mathcal{E} = -\frac{\Delta(N\Phi)}{\Delta t}$$

**Faraday's Law**: the induced e.m.f. is proportional to the rate of change of flux linkage.

**Lenz's Law**: the induced e.m.f. (and hence current) acts in a direction that **opposes** the change in flux that caused it (the minus sign above).

$\mathcal{E}$ = induced e.m.f. (V)
$N$ = number of turns
$\Delta\Phi/\Delta t$ = rate of change of flux (Wb s$^{-1}$ $\equiv$ V)

### Factors Affecting the Induced E.M.F.

- **Rate of change** of flux: faster change $\Rightarrow$ larger e.m.f.
- **Number of turns** $N$: more turns $\Rightarrow$ larger e.m.f.
- **Strength of field** $B$: stronger field $\Rightarrow$ larger flux change.
- **Area** of coil: larger area $\Rightarrow$ more flux.

Lenz's law is a consequence of conservation of energy — the induced current creates a force that opposes the motion causing induction.

### Common Mistake

An e.m.f. is induced only when the flux is **changing**. A conductor stationary in a steady field has zero induced e.m.f., even if the field is strong.

## Formula Summary Sheet

| Formula | Quantity | Units |
|---|---|---|
| $F = BIL\sin\theta$ | Force on current-carrying conductor | N |
| $F = BQv\sin\theta$ | Force on moving charge | N |
| $r = mv/(BQ)$ | Radius of circular orbit in $B$ field | m |
| $V_H = BI/(ntq)$ | Hall voltage | V |
| $v = E/B$ | Velocity selector condition | m s$^{-1}$ |
| $\Phi = BA\cos\theta$ | Magnetic flux | Wb |
| $\mathcal{E} = -\Delta(N\Phi)/\Delta t$ | Faraday's / Lenz's law | V |

**Units:** 1 T = 1 N A$^{-1}$m$^{-1}$; 1 Wb = 1 T m$^2$ = 1 V s
**Right-hand grip rule**: thumb ∥ current, fingers curl in direction of $B$ field.

## Worked Examples

### Example 1 — Force on a Conductor

**Question:** A wire of length 0.15 m carries a current of 3.0 A at 60° to a uniform field of 0.25 T. Calculate the force on the wire.

**Solution**

$$F = BIL\sin\theta = 0.25 \times 3.0 \times 0.15 \times \sin 60^\circ$$
$$F = 0.25 \times 3.0 \times 0.15 \times 0.866 = \mathbf{9.7 \times 10^{-2}}\ \text{N}$$

### Example 2 — Circular Motion of a Charged Particle

**Question:** A proton (mass $1.67 \times 10^{-27}$ kg, charge $1.6 \times 10^{-19}$ C) moves at $2.0 \times 10^6$ m s$^{-1}$ perpendicular to a field of 0.15 T. Calculate the radius of its circular path.

**Solution**

$$r = \frac{mv}{BQ} = \frac{1.67 \times 10^{-27} \times 2.0 \times 10^6}{0.15 \times 1.6 \times 10^{-19}} = \frac{3.34 \times 10^{-21}}{2.40 \times 10^{-20}} = \mathbf{0.14}\ \text{m}$$

### Example 3 — Induced E.M.F.

**Question:** A coil of 200 turns and area 50 cm$^2$ is in a field of 0.30 T perpendicular to the plane of the coil. The field drops to zero in 0.040 s. Calculate the induced e.m.f.

**Solution**

$$\Delta\Phi = B \times A = 0.30 \times 50 \times 10^{-4} = 1.5 \times 10^{-3} \text{ Wb}$$

$$\mathcal{E} = N\frac{\Delta\Phi}{\Delta t} = 200 \times \frac{1.5 \times 10^{-3}}{0.040} = \mathbf{7.5} \text{ V}$$

## Practice Exam Questions

### Section A — Short Answer Questions

**Q1.** Define magnetic flux density and state its SI unit.

*[2 marks]*

**Q2.** A wire of length 8.0 cm is placed perpendicular to a field of flux density 0.40 T and carries a current of 2.5 A. Calculate the force on the wire.

*[2 marks]*

**Q3.** State Faraday's Law and Lenz's Law of electromagnetic induction.

*[3 marks]*

**Q4.** Explain why the speed of a charged particle moving perpendicular to a uniform magnetic field remains constant.

*[2 marks]*

**Q5.** A Hall probe gives a voltage of 3.2 mV when a current of 50 mA passes through a slice of thickness 2.0 mm in a field $B$. Given $n = 8.5 \times 10^{28}$ m$^{-3}$ and $q = 1.6 \times 10^{-19}$ C, calculate $B$.

*[3 marks]*

## Section B — Longer Structured Questions

**Q6.** An electron (mass $9.11 \times 10^{-31}$ kg, charge $1.6 \times 10^{-19}$ C) enters a uniform magnetic field of $2.0 \times 10^{-3}$ T perpendicular to the field with speed $5.0 \times 10^{6}$ m s$^{-1}$.

(a) Calculate the radius of the circular path followed by the electron.

*[2 marks]*

(b) The field strength is doubled. State and explain the effect on the radius.

*[2 marks]*

(c) An electric field is now applied perpendicular to $B$ so that the electron travels in a straight line. Calculate the electric field strength required.

*[2 marks]*

**Q7.** A rectangular coil of 80 turns and dimensions 4.0 cm × 6.0 cm is placed with its plane perpendicular to a uniform field of 0.50 T.

(a) Calculate the flux linkage through the coil.

*[2 marks]*

(b) The coil is rotated so that its plane becomes parallel to the field in 0.030 s. Calculate the mean induced e.m.f.

*[2 marks]*

(c) State and explain the direction of the induced current using Lenz's law.

*[2 marks]*

## Mark Scheme and Answers

**Q1.** Magnetic flux density is the force per unit current per unit length on a wire placed at right angles to the field [1]; unit: tesla (T) or N $A^{-1}$ $m^{-1}$ [1].

**Q2.** $F = BIL\sin 90° = 0.40 \times 2.5 \times 0.080 = \mathbf{0.080}$ N [2].

**Q3.** Faraday's Law: the induced e.m.f. is proportional to the rate of change of flux linkage [1]; $\mathcal{E} = -\Delta(N\Phi)/\Delta t$ [1]. Lenz's Law: the induced e.m.f. acts in a direction to oppose the change in flux that caused it [1].

**Q4.** The magnetic force is always perpendicular to the velocity [1]; a perpendicular force does no work, so kinetic energy and hence speed remain unchanged [1].

**Q5.** $B = V_H ntq/I = (3.2 \times 10^{-3} \times 8.5 \times 10^{28} \times 2.0 \times 10^{-3} \times 1.6 \times 10^{-19})/(50 \times 10^{-3})$ [2] $= \mathbf{0.174}$ T [1].

**Q6(a).** $r = mv/(BQ) = (9.11 \times 10^{-31} \times 5.0 \times 10^{6})/(2.0 \times 10^{-3} \times 1.6 \times 10^{-19})$ [1] $= \mathbf{1.4 \times 10^{-2}}$ m [1].

**Q6(b).** Radius halves [1]; $r = mv/BQ$, so $r \propto 1/B$; doubling $B$ halves $r$ [1].

**Q6(c).** For straight-line motion: $qE = BQv$; $E = Bv = 2.0 \times 10^{-3} \times 5.0 \times 10^{6} = \mathbf{1.0 \times 10^{4}}$ V m$^{\mathbf{-1}}$ [2].

**Q7(a).** $\Phi = BA = 0.50 \times (0.040 \times 0.060) = 1.2 \times 10^{-3}$ Wb [1]; flux linkage $= N\Phi = 80 \times 1.2 \times 10^{-3} = \mathbf{9.6 \times 10^{-2}}$ Wb-turns [1].

**Q7(b).** $\Delta(N\Phi) = 9.6 \times 10^{-2}$ Wb-turns (falls to zero) [1]; $\mathcal{E} = 9.6 \times 10^{-2}/0.030 = \mathbf{3.2}$ V [1].

**Q7(c).** By Lenz's law the induced current opposes the decrease in flux [1]; the current flows in the direction that would create a field to oppose the rotation / maintain the flux through the coil [1].

## Revision Checklist

Use this checklist to track your understanding. Tick each box when you are confident:

| | Learning Objective | Confidence (1–3) |
|---|---|---|
| ☐ | Define magnetic flux density; use $F = BIL\sin\theta$ | |
| ☐ | Apply Fleming's left-hand rule to find force directions | |
| ☐ | Use $F = BQv\sin\theta$ for force on a moving charge | |
| ☐ | Derive and use $r = mv/(BQ)$ for circular orbit radius | |
| ☐ | Explain and use the Hall effect; use $V_H = BI/(ntq)$ | |
| ☐ | Explain the velocity selector condition $v = E/B$ | |
| ☐ | Sketch field patterns for straight wire, circular coil and solenoid | |
| ☐ | Apply the right-hand grip rule for field direction around a wire | |
| ☐ | Explain forces between parallel current-carrying conductors | |
| ☐ | Define magnetic flux $\Phi = BA\cos\theta$ and flux linkage $N\Phi$ | |
| ☐ | State and apply Faraday's and Lenz's laws | |
| ☐ | Identify factors that affect the magnitude of induced e.m.f. | |

*Key: 1 = Need more work 2 = Getting there 3 = Confident*

**Good luck with your revision!**

Faraday's law — one equation — explains the generator, the transformer, and the electric motor. Master flux linkage and you understand how almost all electrical power is generated.